

1) Poisson Process

Definition of Poisson Process PP:

$$X(t) \sim \text{Poi}(\lambda t) \implies \begin{aligned} P[X(t) = k] &= \frac{(\lambda t)^k e^{-\lambda t}}{k!}, \quad k \geq 0 \\ \mathbb{E}[X(t)] &= \lambda t \\ \text{var}(X(t)) &= \lambda t \end{aligned}$$

Moreover

$$X(t_1) - X(t_2) \sim \text{Poi}((t_1 - t_2)\lambda), \quad t_1 > t_2$$

A fundamental property of a Poisson process is that the number of events occurring in non-overlapping intervals are independent.

$$P[A|B] = P[A] \iff A \perp B$$

Recall:

$$P[A|B] = \frac{P[A, B]}{P[B]}$$

In the exercises, this is used only when:

- one PP conditioned on one past or future event = One PP with max 2 sampled points

Total distribution is used

- One PP with at least 3 sampled points

This total probability formula is useful:

$$P(E | C) = \sum_{k \in \mathcal{K}} P(E | B_k \cap C) \cdot P(B_k | C)$$

Practical example:

$$\begin{aligned} P[X_1(3) = 2 | X_1(2) + X_2(2) = 1] &= \sum_{k=0}^1 P[X_1(3) = 2, X_1(2) = k | X_1(2) + X_2(2) = 1] \\ &= \sum_k \frac{P[X_1(3) = 2, X_1(2) = k, X_1(2) + X_2(2) = 1]}{P[X_1(2) + X_2(2) = 1]} \\ &= \sum_{k=0}^1 P[X_1(3) = 2 | X_1(2) = k, X_1(2) + X_2(2) = 1] P[X_1(2) = k | X_1(2) + X_2(2) = 1] \end{aligned}$$

The first part becomes:

$$\begin{aligned} P[X_1(3) - X_1(2) = 2 - k | X_1(2) + X_2(2) = 1 - k] \\ = P[X_1(1) = 2 - k | X_2(2) = 1 - k] = P[X_1(1) = 2 - k] \end{aligned}$$

The second becomes (its sum over all k must be 1)

$$\frac{P[X_1(2) = k, X_1(2) + X_2(2) = 1]}{P[X_1(2) + X_2(2) = 1]} = \frac{P[X_1(2) = k] P[X_2(2) = 1 - k]}{P[X_1(2) + X_2(2) = 1]}$$

2) Poisson Process Queues

In a queue we have the arrival distribution A and service time S .

Usually we have $A(t) \sim \text{Poi}(\lambda t)$ and S either uniform, constant or exponentially distributed.

$A(t)$ is the cumulative arrivals up to t

$N(t)$ are the users in service at time t

From here we define the number of customers in the system at time t as $N(t)$.

If there are infinite queues/capacity, then $N(t) \sim \text{Poi}(\lambda_p(t))$ with rate

$$\lambda_p(t) = \lambda \int_0^t 1 - G(z) dz \xrightarrow{t \rightarrow \infty} \lambda E[S]$$

Where $G(z)$ is the CDF of the Service time. Notice that $t \rightarrow \infty$ doesn't need to be the limit, just $t >$ then when the service time CDF converges to 1.

But notice that

$$P[N(t) = k] = \frac{[\lambda_p(t)]^k e^{-\lambda_p(t)}}{k!}$$

It is possible to find the probability of having n customers in service at t with k arrivals before using this:

- Constant service time β :

$$\begin{aligned} P[N(t) = n | A(t) = k] &= \frac{P[N(t) = n, A(t) = k]}{P[A(t) = k]} = \frac{P[A(t - \beta) = k - n] P[A(\beta) = n]}{P[A(t) = k]} \\ &= \binom{k}{n} \left(\frac{\beta}{t}\right)^n \left(\frac{t - \beta}{t}\right)^{k-n} \\ &\sim \text{Binomial}\left(k, \frac{\beta}{t}\right) \end{aligned}$$

- Random service time:

$$\begin{aligned} P[N(t) = n | A(t) = k] &= \binom{k}{n} \left[\frac{\lambda_p(t)}{\lambda t}\right]^n \left[1 - \frac{\lambda_p(t)}{\lambda t}\right]^{k-n}, \quad 0 \leq n \leq k \\ &\sim \text{Binomial}\left(k, \frac{\lambda_p(t)}{\lambda t}\right) \end{aligned}$$

Most importantly, the constant service time case is a special case of the random service time.

Given $A(t) = k$, the arrival epochs are distributed as **order statistics of k i.i.d. Uniform(0, t) random variables**.

The probability that an arrival at time $U \sim \text{Unif}(0, t)$ is still in service at t is:

$$p_t = \frac{1}{t} \int_0^t (1 - G(t - u)) du = \frac{\lambda_p(t)}{\lambda t}$$

3) MC

3.1) Analysis of MC:

- Diagram
 - Classes
 - Recurrence/transience
 - Periodicity
-

3.2) Probability absorption into class a :

If the transient state is only one ($|C_t| = 1$): $P[\text{go to class } a] / P[\text{leave trans state}]$

If the transient state is a class with $|C_t| > 1$ then first step analysis is required:

Define: $\mu_i = P[\text{go to } a \text{ starting from } i]$

$$\mu_i = \sum_j P_{ij} \mu_j$$

Boundary condition: $\mu_a = 1 \quad \mu_b = 0, \quad b \in \{\text{all other abs states}\}$

where the boundary condition is a direct result of the summation.

3.3) Compute Steady State Probabilities:

For each irreducible **positive** recurrent class solve:

$$\begin{cases} \pi_C P_C = \pi_C \\ \sum_{\pi_i \in \pi_C} \pi_i = 1 \end{cases}$$

For each irreducible transient or negative recurrent class it is 0.

3.4) Compute Avg Recurrence Times:

For each irreducible **positive** recurrent class solve:

$$m_i = \frac{1}{\pi_i}$$

For each irreducible transient or negative recurrent class it is ∞ .

3.5) Avg Number of Visits starting from a

Starting from a inside a pos rec chain:

$$W_{a\bullet}^{(n)} = \sum_{k=0}^{n-1} [P^k]_{a\bullet} \stackrel{n \rightarrow \infty}{\approx} n\pi_{\bullet}$$

Notice that this can be easily computed as:

Set

$$X_0 = (0, \dots, 0, \underbrace{1}_{a\text{-th pos}}, 0, \dots, 0)$$

Find $X_k = X_{k-1}P$, $k = 1, \dots, n-1$ (where X_1 is the a -th row of P and X_0 is a null row where only a -th entry is 1).

$$X_k = X_{k-1}P = X_0P^k = [P^k]_{a\bullet}$$

Then the $W_{a\bullet}^{(n)}$ matrix is just the sum, to find j just select the j -th entry

$$W_{a\bullet}^{(n)} = \sum_{k=0}^{n-1} X_k \xrightarrow{n \rightarrow \infty} n\pi_{\bullet}$$

Starting from a inside transient state $\mathcal{T}$:

Same as before for finite n

The limit is a bit different:

- If the target state is transient

$$W_{ij}^{(\infty)} = \delta_{ij} + \sum_{k \in \mathcal{T}} P_{ik} W_{kj}^{(\infty)} \quad \text{for every } i \in \mathcal{T}$$

where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise.

- If the target is in recurrent state then as before

Finally notice that

$$k_i = \sum_{j \in \mathcal{T}} W_{ij}$$

3.6) Compute Mean and Variance of the first passage time to state a

Let k_i^a = mean time to reach a from i

Do one system of this form for each required a

Clearly $k_a^a = 0$ however, for the final result consider $k_a^a = 1/\pi_a$

$$k_i^a = 1 + \sum_{j \neq a} P_{ij} k_j^a$$

If $i \nrightarrow a$, or if $i \rightarrow a$ but the chain can be trapped in another closed class disjoint from a , then $k_i^a = \infty$

Let α_i^a = secon moment of time to reach a from i

Once the means k_i^a are known just build the following system

$$\alpha_i = 1 + \sum_{j \neq a} P_{ij} (2k_j^a + \alpha_j)$$

Finally the variance is

$$\alpha_i - k_i^2$$

3.7) Computing $\lim_{n \rightarrow \infty} P^n$:

- Doesn't exist for periodic classes "X"
- Is null for transitions (from transient) into transient states
- Doesn't exist for transitions (from transient) into periodic classes "X"

For Positive Recurrent Aperiodic Class it is just the steady state probability:

For a submatrix P_{C_i} where C_i is a pos. recurrent aperiodic class the rows will be

$$\text{row } i: (\pi_1 \quad \pi_2 \quad \dots \quad \pi_n)$$

For transitions (from transient) to pos recurrent aperiodic it is the abs probability times steady state of the state it transitions into

$$P(\text{absorbed in } C_i) \cdot \pi_n \quad \text{for all entries}$$

3.8) Computing $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n P^k$:

- Exists *always*
- If standard limit exists, it is equal to it

For Positive Recurrent Periodic Class it is just the steady state probability (same as before for aperiodic)

For For transitions (from transient) to pos recurrent periodic it is the abs probability times steady state of the state it transitions into (same as before fro transient $\rightarrow$ aperiodic)

Transition Type (i $\rightarrow$ j)	Class of i	Class of j	Limit $\lim_{n \rightarrow \infty} P_{ij}^n$
To Transient	Any	Transient	0
Across Recurrent	Recurrent C_1	Recurrent C_2 ($C_1 \neq C_2$)	0
Within Aperiodic Recurrent	Recurrent C_i	Same C_i (Aperiodic)	π_j (from $\pi_{C_i} P_{C_i} = \pi_{C_i}, \sum \pi = 1$)
Within Periodic Recurrent	Recurrent C_i	Same C_i (Periodic)	Does not exist (X)
Transient $\rightarrow$ Aperiodic Recurrent	Transient	Recurrent C_i (Aperiodic)	$P(\text{absorbed in } C_i \mid X_0 = i) \cdot \pi_j$
Transient $\rightarrow$ Periodic Recurrent	Transient	Recurrent C_i (Periodic)	0 if unreachable; Does not exist (X) if reachable and oscillating

3.9) Mean recurrence time (avg return times)

For pos recurrent state $\mu_i = 1/\pi_i$

For transient or neg rec states $\mu_i = \infty$

4) First Step Analysis

General formula:

$$W_{ij} = \mathbb{I}\{\overset{\text{yellow}}{i} = \overset{\text{red}}{j}\} + \sum_k P_{\overset{\text{yellow}}{i} \overset{\text{blue}}{k}} W_{\overset{\text{blue}}{k} \overset{\text{red}}{j}}$$

This is used for

- avg visits
- passage times

5) GBN

If P is known, then also π_G, π_B are known by solving $\pi = \pi P$ with $\pi_G + \pi_B = 1$.

If π_G, π_B is known P cannot be calculated. If also $\mathbb{E}[\text{consecutive good/bad slots}]$ is known (one of the two) then

$$p_{GB} = \frac{1}{\mathbb{E}[\text{consec good}]} \quad p_{BG} = \frac{1}{\mathbb{E}[\text{consec bad}]}$$

From here a parametric P can be defined and by solving $\pi = P\pi$ with $a + b = 1$ as an additional condition the whole P is found.

If both are known it is directly found by recalling that each row = 1.

If the throughput of no protocol is asked:

$$\eta = \pi_G P[\text{success}|G] + \pi_B P[\text{success}|B]$$

A GBN on a two state MC works as follows. It has RTT of m slots

- Normal GBN with iid error ϵ on forward channel and m slots to return

$$\eta = \frac{1 - \epsilon}{1 - \epsilon + m\epsilon}$$

(where $P\{BG\} = 1 - \epsilon$ since iid and $P\{GB\} = \epsilon$)

- Error Free feedback throughput:

$$\eta = \frac{P_{10}^{(m)}}{P_{10}^{(m)} + mP_{01}} = \frac{P_{BG}^{(m)}}{P_{BG}^{(m)} + mP_{GB}}$$

- Feedback channel with iid error δ :

$$\eta = \frac{(1 - \delta)P_{BG}^{(m)}}{(1 + (m - 1)\delta)P_{BG}^{(m)} + m((1 - \delta)P_{GB} + \delta P_{GB}^{(m)})}$$

If iid errors on both feedback and forward then use the last provided formula by recalling $P_{BG}^{(2)} = P_{BG} = 1 - \epsilon$ since iid and $P_{GB} = \epsilon$

6) Min/Max of two distributions

Let A, B be two non negative independent rvs, then define $W = \min(A, B)$

$$W = \min(A, B) > t \iff A > t \cap B > t \longrightarrow P[W > t] = P[A > t]P[B > t]$$

Sometimes this is enough to know what the distribution is.

Calculate $\mathbb{E}[W]$:

$$\mathbb{E}[W] = \int_0^\infty P(W > t) dt$$

Recall also:

$$F_W(t) = 1 - (1 - F_A(t))(1 - F_B(t))$$

Specific cases:

- $A \sim \text{Exp}(\lambda_A) \ B \sim \text{Exp}(\lambda_B)$:

$$\begin{aligned} W &\sim \text{Exp}(\lambda_A + \lambda_B) \\ \mathbb{E}[W] &= \frac{1}{\lambda_A + \lambda_B} \end{aligned}$$

- $A \sim \text{Exp}(\lambda) \ B \sim \text{Constant}(T)$

$$\begin{aligned} W &\sim \text{Truncated Exp}(\lambda) \text{ at } T \\ \mathbb{E}[W] &= \frac{1 - e^{-\lambda T}}{\lambda} \end{aligned}$$

In general a uniform rv truncates the other rv.

Now notice:

$$\max(A, B) + \min(A, B) = A + B \rightarrow \max(A, B) = A + B - \min(A, B)$$

Therefore the expectation of the max is very easily obtained:

$$\mathbb{E}[\max(A, B)] = \mathbb{E}[A] + \mathbb{E}[B] - \mathbb{E}[W]$$

So the distribution of max can be found by finding the distribution of min:

$$\begin{aligned} P[\max(A, B) > t] &= P[A > t] + P[B > t] - P[\min(A, B) > t] \\ &= P[A > t] + P[B > t] - P[A > t]P[B > t] \end{aligned}$$

7) Renewal Reward

Important difference:

"Per visit" / "Sojourn time" / "Holding time": Conditioned on entering the state $\implies \mu_R = \gamma T$.

"Per cycle": Unconditioned over the full loop, weighting the probability of entering the state $\implies E[T_R] = \alpha \gamma T$.

$$E[T_{\text{between two events } i}] = \frac{E[\text{short cycle}]}{P_i}$$

7.1) Two ON/OFF Phases

Let a system cycle between two phases of expected duration T_1, T_2 that produce r_i

Probability to be in state i

$$p_i = \frac{\mathbb{E}[T_i]}{\mathbb{E}[T_1] + \mathbb{E}[T_2]}$$

Rate:

$$\eta = r_1 p_1 + r_2 p_2$$

If there are multiple independent processes with ON/OFF phases then

Find p_i, η_i for every process

then

$$\eta = \sum_{k=0}^n P(k \text{ are working}) \cdot r(k)$$

7.2) Semi Markov Process

≥ 3 states, the *order* of visits is random (governed by probabilities, not a fixed cycle), and/or a state has **competing exit clocks**.

First find embedded matrix, then find π_i

Then the mean times:

- Single exit clock:

$$\mu_i = E[T_i] = \sum_j P_{ij} \tau_{ij}$$

where $\tau_{ij} = \mathbb{E}[\text{time in state } i \mid \text{next state is } j]$.

- Competing exponential clocks:

$$P[T_1 < T_2] = \frac{\lambda_1}{\lambda_1 + \lambda_2} \quad \min \sim \text{Exp}[\lambda_1 + \lambda_2]$$

- Exponential vs deterministic:

$$P[\text{Exp}(\lambda) < c] = 1 - e^{-\lambda c}, \quad \mathbb{E}[\min(\text{Exp}(\lambda), c)] = \frac{1 - e^{-\lambda c}}{\lambda}$$

Finally un normalize

$$p_i = \frac{\pi_i \mu_i}{\sum_k \pi_k \mu_k}$$

7.3) Regenerative Analysis

you need throughput, a blocking/rejection fraction, or an average delay in a system that isn't a clean birth-death CTMC (thresholds, timeouts, synchronized departures).

Find the start of cycle
 Then decompose the cycle:
 Then compute
 Throughput:

$$E[\text{data sent per cycle}] / E[C]$$

Rejection fraction:

$$E[\text{\#rejected per cycle}] / E[\text{\#offered per cycle}]$$

both share the same $\mathbb{E}[C]$, so it cancels; you only need the two expected counts. Note that for a Poisson arrival process with rate λ , $\mathbb{E}[\text{\# offered per cycle}] = \lambda \mathbb{E}[C]$.

Fraction of time empty:

$$E[\text{idle sub-phase}] / E[C]$$

Average Delay ($\mathbb{E}[T]$):

$$L = \frac{\mathbb{E} \left[\int_0^C N(t) dt \right]}{\mathbb{E}[C]}$$

$$\lambda_{\text{eff}} = \frac{\mathbb{E}[\text{admitted packets per cycle}]}{\mathbb{E}[C]}$$

$$\mathbb{E}[T] = \frac{L}{\lambda_{\text{eff}}} = \frac{\mathbb{E} \left[\int_0^C N(t) dt \right]}{\mathbb{E}[\text{admitted packets per cycle}]}$$

8) Other stuff

8.1) Independent and Identically Distributed (i.i.d.)

If every process has the same success probability $p_i = p$, the total number of operational processes follows a standard **Binomial distribution**, $K \sim \text{Binomial}(n, p)$:

Probability that exactly k processes are working:

$$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k \in \{0, 1, \dots, n\}$$

All n are working ($k = n$):

$$P(K = n) = p^n$$

All n are off ($k = 0$):

$$P(K = 0) = (1 - p)^n$$

Exactly one is working ($k = 1$):

$$P(K = 1) = np(1 - p)^{n-1}$$

At least one is working:

$$P(K \geq 1) = 1 - P(K = 0) = 1 - (1 - p)^n$$

8.2) CSMA/Slotted Aloha

If each successful transmission brings gain G and each blocked/failed attempt costs C :

- Total attempts rate = λ_{total} .
- Success rate = $\lambda_{\text{succ}} = \lambda_{\text{new}}$ (if all packets eventually succeed).
- Failure rate = $\lambda_{\text{total}} - \lambda_{\text{succ}}$.
- Net utility rate:

$$\text{Gain Rate} = G \cdot \lambda_{\text{succ}} - C \cdot (\lambda_{\text{total}} - \lambda_{\text{succ}})$$

8.3) Example of Avg visits

$$P = \begin{pmatrix} P_{00} & P_{01} & P_{02} \\ P_{10} & P_{11} & P_{12} \\ 0 & 0 & 1 \end{pmatrix}$$

$$W_{ij}^{(\infty)} = \delta_{ij} + \sum_{k \in \mathcal{T}} P_{ik} W_{kj}^{(\infty)}$$

$j = 0$

$$\begin{cases} W_{00}^{(\infty)} = 1 + P_{00} W_{00}^{(\infty)} + P_{01} W_{10}^{(\infty)} \\ W_{10}^{(\infty)} = 0 + P_{10} W_{00}^{(\infty)} + P_{11} W_{10}^{(\infty)} \end{cases}$$

$j = 1$

$$\begin{cases} W_{01}^{(\infty)} = 0 + P_{00} W_{01}^{(\infty)} + P_{01} W_{11}^{(\infty)} \\ W_{11}^{(\infty)} = 1 + P_{10} W_{01}^{(\infty)} + P_{11} W_{11}^{(\infty)} \end{cases}$$